INCH-POUND

MIL-C-39022/9D
25 May 2001
SUPERSEDING
MIL-C-39022/9C
31 October 1973

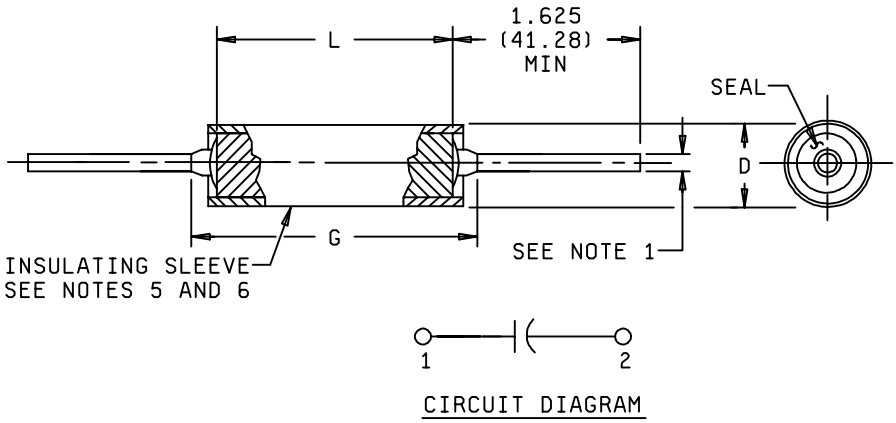
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED, PLASTIC FILM DIELECTRIC, DIRECT AND ALTERNATING CURRENT (HERMETICALLY SEALED IN METAL CASES) ESTABLISHED RELIABILITY, STYLE CHR01A, CHR01B, CHR01C, CHR01D, CHR01E, CHR01F, CHR01G, CHR01H, CHR01J, CHR01K, CHR01L, CHR01M, CHR01N, CHR01P, AND CHR01R (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39022.

INACTIVE FOR NEW DESIGN AFTER 31 OCTOBER 1973.
FOR NEW DESIGN, USE MIL-PRF-83421.



CIRCUIT DIAGRAM

NOTES:
1. Number 22 AWG wire (.025 ± .002 inch (0.64 ± 0.05 mm)) for case diameters of .312 inch (7.92 mm) and less. Number 20 AWG wire (.032 ± .002 inch (0.81 ± 0.05 mm)) for case diameters of .400 inch (10.16 mm) and .500 (12.70 mm).
   Number 18 AWG wire (.040 inch ± .002 inch (1.02 ± 0.05 mm)) for case diameters of .562 inch (14.27 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 inch (0.79 mm) on either end. Insulating sleeve thickness shall not exceed .005 inch (0.13 mm).
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.
8. Lead length may be a minimum of one inch long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic. Seal headers may be magnetic.

Sleeves: Heat shrink only.

Leads:

500 inch (12.70 mm) and smaller case diameters: Solder-coated copper clad steel. Type N2 of MIL-STD-1276 may be furnished when specified in the contract or purchase order (see MIL-PRF-39022 ordering data).

.562 inch (14.27 mm) and larger case diameters: Type C of MIL-STD-1276.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Rated voltage: DC and ac (see table I).

Operating temperature range: -55°C to +100°C; linearly derate to 50 percent of rated voltage for +125°C operating.

Failure rate level: M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent) (established at dc rated voltage at +100°C.

Flash point of impregnant: Not applicable.

Radiographic inspection (X-ray): Not applicable.

Thermal shock: Method 107 of MIL-STD-202, condition B.

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIA; no evidence of leakage with the exception of the following:

Styles	lbf/in^2	Exposure time hours	Maximum dwell hours	R1 Reject limit ($atm\ cm^3/s$)
CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, AND CHR01P	45 ± 5	1 +1-0	1	1×10^{-6}
CHR01C, CHR01F, CHR01J, CHR01M, AND CHR01R	45 ± 5	1 +1-0	1	1×10^{-5}

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TABLE I. Electrical characteristics, dimensions, and dash numbers.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01A, CHR01B, AND CHR01C - 30 VOLTS (DC RATING)															
0.001	.500 (12.70)	.170 (4.32)	.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	22.0	.001	22.0	.001	22.0	.006
0.0012				-007-	-008-	-009-	-010-	-011-	-012-						.007
0.0015				-013-	-014-	-015-	-016-	-017-	-018-						.008
0.0018				-019-	-020-	-021-	-022-	-023-	-024-						.010
0.002				-025-	-026-	-027-	-028-	-029-	-030-						.011
0.0022				-031-	-032-	-033-	-034-	-035-	-036-						.012
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.002		.015
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.002		.020
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.003		.026
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.003		.028
0.0068				-061-	-062-	-063-	-064-	-065-	-066-				.004		.037
0.0082				-067-	-068-	-069-	-070-	-071-	-072-				.005		.045
0.01				-073-	-074-	-075-	-076-	-077-	-078-				.006		.055
0.012				-079-	-080-	-081-	-082-	-083-	-084-				.007		.068
0.015				-085-	-086-	-087-	-088-	-089-	-090-				.008		.082
0.018				-091-	-092-	-093-	-094-	-095-	-096-				.010		.100
0.02				-097-	-098-	-099-	-100-	-101-	-102-				.011		.110
0.022				-103-	-104-	-105-	-106-	-107-	-108-				.012		.121
0.027	.562 (14.27)		.762 (19.35)	-109-	-110-	-111-	-112-	-113-	-114-				.015		.149
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.002		.018		.182
0.039				-301-	-302-	-303-	-304-	-305-	-306-		.002		.021		.215
0.047				-121-	-122-	-123-	-124-	-125-	-126-		.003		.026		.259
0.05				-127-	-128-	-129-	-130-	-131-	-132-		.003		.028		.275
0.056				-307-	-308-	-309-	-310-	-311-	-312-		.003		.031		.310
0.068	.687 (17.45)		.887 (22.53)	-133-	-134-	-135-	-136-	-137-	-138-		.004		.040		.374
0.082				-139-	-140-	-141-	-142-	-143-	-144-		.005		.050		.451
0.1		.193 (4.90)		-145-	-146-	-147-	-148-	-149-	-150-		.006		.055		.550
0.12		.193 (4.90)		-151-	-152-	-153-	-154-	-155-	-156-		.007		.068		.661
0.15	.562 (14.27)	.235 (5.97)	.762 (19.35)	-157-	-158-	-159-	-160-	-161-	-162-	22.0	.008	22.0	.082	22.0	.824
0.20	.687 (17.45)		.887 (22.53)	-163-	-164-	-165-	-166-	-167-	-168-		.011		.110	20.0	1.000
0.22				-169-	-170-	-172-	-172-	-173-	-174-		.012		.121	18.5	1.017

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01A, CHR01B, AND CHR01C - 30 VOLTS (DC RATING)															
0.27				-175-	-176-	-177-	-178-	-179-	-180-		.015		.149	17.0	1.148
0.33				-181-	-182-	-183-	-184-	-185-	-186-		.018		.182	16.0	1.322
0.39				-313-	-314-	-315-	-316-	-317-	-318-		.021		.215	15.0	1.463
0.47		.312 (7.92)		-187-	-188-	-189-	-190-	-191-	-192-		.026		.259	14.0	1.645
0.50				-193-	-194-	-195-	-196-	-197-	-198-		.028		.275	13.5	1.688
0.68				-199-	-200-	-201-	-202-	-203-	-204-		.037		.374	12.0	2.041
0.82	.813 (20.65)		1.013 (25.73)	-205-	-206-	-207-	-208-	-209-	-210-		.045		.451	10.0	2.049
1.0				-211-	-212-	-213-	-214-	-215-	-216-		.055		.550	8.4	2.100
1.5		.400 (10.16)		-217-	-218-	-219-	-220-	-221-	-222-		.082		.824	5.8	2.172
2.0				-223-	-224-	-225-	-226-	-227-	-228-		.110		1.100	4.4	2.200
2.2				-229-	-230-	-231-	-232-	-233-	-234-		.121		1.210	4.0	
3.0	1.063 (27.00)		1.263 (32.08)	-235-	-236-	-237-	-238-	-239-	-240-		.165		1.654	2.9	
3.3	1.063 (27.00)		1.263 (32.08)	-241-	-242-	-243-	-244-	-245-	-246-		.182		1.818	2.7	
4.0	1.375 (34.93)		1.575 (40.01)	-247-	-248-	-249-	-250-	-251-	-252-		.220		2.200	2.2	
4.7		.500 (12.70)		-253-	-254-	-255-	-256-	-257-	-258-		.259	18.7		1.87	
5.0		.500 (12.70)		-259-	-260-	-261-	-262-	-263-	-264-		.275	17.6		1.76	
6.8		.562 (14.27)		-319-	-320-	-321-	-322-	-323-	-324-		.374	13.0		1.30	
8.0				-325-	-326-	-327-	-328-	-329-	-330-		.440	11.0		1.10	
10.0				-331-	-332-	-333-	-334-	-335-	-336-		.550	8.8		.88	
15.0	1.875 (47.83)		2.075 (52.71)	-337-	-338-	-339-	-340-	-341-	-342-		.824	5.6		.56	
20.0		.670 (17.02)		-343-	-344-	-345-	-346-	-347-	-348-		1.100	4.4		.44	
22.0		.670 (17.02)		-349-	-350-	-351-	-352-	-353-	-354-		1.210	4.0		.40	
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS (DC RATING)															
0.001	.500 (12.70)	.170 (4.32)	.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	36.0	.001	36.0	.001	36.0	.009
0.0012				-007-	-008-	-009-	-010-	-011-	-012-						.012
0.0015				-013-	-014-	-015-	-016-	-017-	-018-						.014
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.002		.016
0.002				-025-	-026-	-027-	-028-	-029-	-030-						.018
0.0022				-031-	-032-	-033-	-034-	-035-	-036-						.020
0.0027				-037-	-038-	-039-	-040-	-041-	-042-						.020
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.003		.030

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS (DC RATING)															
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.004		.042
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.005		.045
0.0068				-061-	-062-	-063-	-064-	-065-	-066-				.006		.061
0.0082	.562 (14.27)		.762 (19.35)	-067-	-068-	-069-	-070-	-071-	-072-				.007		.074
0.01				-073-	-074-	-075-	-076-	-077-	-078-				.009		.090
0.012				-079-	-080-	-081-	-082-	-083-	-084-				.011		.108
0.015				-085-	-086-	-087-	-088-	-089-	-090-				.013		.135
0.018		.193 (4.90)		-091-	-092-	-093-	-094-	-095-	-096-		.002		.016		.163
0.02				-097-	-098-	-099-	-100-	-101-	-102-				.018		.180
0.022				-103-	-104-	-105-	-106-	-107-	-108-				.020		.121
0.027				-109-	-110-	-111-	-112-	-113-	-114-				.024		.243
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.003		.030		.296
0.039	.687 (17.45)	.170 (4.32)	.887 (22.53)	-283-	-284-	-285-	-286-	-287-	-288-		.004		.035		.351
0.047				-121-	-122-	-123-	-124-	-125-	-126-		.004		.040		.423
0.05				-127-	-128-	-129-	-130-	-131-	-132-		.005		.045		.450
0.056		.193 (4.90)		-289-	-290-	-291-	-292-	-293-	-294-		.005		.050		.504
0.068				-133-	-134-	-135-	-136-	-137-	-138-		.006		.060	34.0	.578
0.082	.813 (20.65)		1.013(25.73)	-139-	-140-	-141-	-142-	-143-	-144-		.007		.070	32.0	.656
0.1	.813 (20.65)		1.013(25.73)	-145-	-146-	-147-	-148-	-149-	-150-		.009		.090	30.0	.750
0.12	.687 (17.45)	.235 (5.97)	.887 (22.53)	-151	-152-	-153-	-154	-155-	-156-		.011		.108	30.0	.901
0.15	.687 (17.45)	.235 (5.97)	.887 (22.53)	-157-	-158-	-159-	-160-	-161-	-162-	36.0	.013	36.0	.135	26.0	.974
0.20	.813 (20.65)		1.013 (25.73)	-163-	-164-	-165-	-166-	-167-	-168-		.018		.180	24.0	1.200
0.22	.813 (20.65)		1.013 (25.73)	-169-	-170-	-171-	-172-	-173-	-174-		.020		.198	20.0	1.100
0.27	.687 (17.45)	.312 (7.92)	.887 (22.53)	-175-	-176-	-177-	-178-	-179-	-180-		.024		.243	19.0	1.282
0.33	.687 (17.45)		.887 (22.53)	-181-	-182-	-183-	-184-	-185-	-186-		.030		.298	18.0	1.488
0.39	8.13 (20.65)		1.013 (25.73)	-295-	-296-	-297-	-298-	-299-	-300-		.035		.351	17.0	1.658
0.47				-187-	-188-	-189-	-190-	-191-	-192-		.042		.423	16.0	1.880
0.50		.400 (10.16)		-193-	-194-	-195-	-196-	-197-	-198-		.045		.450	15.0	1.875
0.68				-199-	-200-	-201-	-202-	-203-	-204-		.062		.612	14.0	2.381
0.82	1.063(27.00)		1.263 (32.08)	-205-	-206-	-207-	-208-	-209-	-210-		.074		.738	12.0	2.459
1.0				-211-	-212-	-213-	-214-	-215-	-216-		.090		.900	10.0	2.500
1.5				-217-	-218-	-219-	-220-	-221-	-222-		.135		1.348	8.0	2.996

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS (DC RATING)															
2.0	1.125 (28.58)	.500 (12.70)	1.325 (33.66)	-223-	-224-	-225-	-226-	-227-	-228-		.180		1.800	6.0	3.000
2.2	1.125 (28.58)		1.325 (33.66)	-229-	-230-	-231-	-232-	-233-	-234-		.198		1.980	5.8	3.190
3.0	1.375 (34.93)		1.575 (40.01)	-235-	-236-	-237-	-238-	-239-	-240-		.271		2.707	4.5	3.384
3.3				-241-	-242-	-243-	-244-	-245-	-246-		.298		2.975	4.1	3.389
4.0		.562 (14.27)		-301-	-302-	-303-	-304-	-305-	-306-		.360		3.600	3.6	3.600
4.7		.670 (17.02)		-307-	-308-	-309-	-310-	-311-	-312-		.423	30.0		3.2	
5.0				-313-	-314-	-315-	-316-	-317-	-318-		.450	28.8		3.0	
6.8	1.875 (47.83)		2.075 (52.71)	-319-	-320-	-321-	-322-	-323-	-324-		.612	21.2		2.2	
8.0				-325-	-326-	-327-	-328-	-329-	-330-		.720	18.0		1.8	
10.0				-331-	-332-	-333-	-334-	-335-	-336-		.900	14.4		1.5	
STYLES CHR01G, CHR01H, AND CHR01J - 100 VOLTS (DC RATING)															
0.001	.500 (12.70)	.170 (4.32)	.700 (17.78)	-001-	-002-	-003-	-004-	-005-	-006-	60.0	.001	60.0	.002	60.0	.015
0.0012				-007-	-008-	-009-	-010-	-011-	-012-						.018
0.0015				-013-	-014-	-015-	-016-	-017-	-018-						.022
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.003		.027
0.002				-025-	-026-	-027-	-028-	-029-	-030-						.030
0.0022				-031-	-032-	-033-	-034-	-035-	-036-						.033
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.004		.041
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.005		.050
0.0047				-049-	-050-	-051-	-052-	-053-	-054-				.007		.071
0.005				-055-	-056-	-057-	-058-	-059-	-060-				.008		.075
0.0068	.687 (17.45)		.887 (22.53)	-061-	-062-	-063-	-064-	-065-	-066-				.010		.102
0.0082				-067-	-068-	-069-	-070-	-071-	-072-				.012		.123
0.01				-073-	-074-	-075-	-076-	-077-	-078-		.002		.015		.150
0.012				-079-	-080-	-081-	-082-	-083-	-084-				.018	59.0	.177
0.015				-085-	-086-	-087-	-088-	-089-	-090-				.022	58.0	.217
0.018		.193 (4.90)		-091-	-092-	-093-	-094-	-095-	-096-		.003		.027	57.0	.257
0.02				-097-	-098-	-099-	-100-	-101-	-102-				.030	55.0	.275
0.022				-103-	-104-	-105-	-106-	-107-	-108-				.033	53.0	.292
0.027				-109-	-110-	-111-	-112-	-113-	-114-		.004		.041	51.0	.344
0.033				-115-	-116-	-117-	-118-	-119-	-120-		.005		.049	50.0	.413
0.039		.235 (5.97)		-121-	-122-	-123-	-124-	-125-	-126-		.006		.059	48.0	.468

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01G, CHR01H, AND CHR01J - 100 VOLTS (DC RATING)															
0.047				-127-	-128-	-129-	-130-	-131-	-132-		.007		.071	47.0	.552
0.050				-133-	-134-	-135-	-136-	-137-	-138-		.008		.075	46.0	.575
0.056				-139-	-140-	-141-	-142-	-143-	-144-		.008		.084	46.0	.644
0.068	.813 (20.65)		1.013(25.73)	-145-	-146-	-147-	-148-	-149-	-150-		.010		.102	42.0	.714
0.082	.687 (17.45)	.312 (7.92)	.887 (22.53)	-151-	-152-	-153-	-154-	-155-	-156-	60.0	.012	60.0	.123	38.0	.779
0.10				-157-	-158-	-159-	-160-	-161-	-162-		.015		.150	36.0	.990
0.12				-163-	-164-	-165-	-166-	-167-	-168-		.018		.180	35.0	1.051
0.15	.813 (20.65)		1.013 (25.73)	-169-	-170-	-171-	-172-	-173-	-174-		.022		.225	31.0	1.160
0.20				-175-	-176-	-177-	-178-	-179-	-180-		.030		.300	29.0	1.450
0.22				-181-	-182-	-183-	-184-	-185-	-186-		.033		.330	25.0	1.375
0.27	1.063 (27.00)		1.263 (32.08)	-187-	-188-	-189-	-190-	-191-	-192-		.041		.405	24.0	1.621
0.33				-193-	-194-	-195-	-196-	-197-	-198-		.050		.496	23.0	1.901
0.39		.400 (10.16)		-199-	-200-	-201-	-202-	-203-	-204-		.059		.585	22.0	2.145
0.47				-205-	-206-	-207-	-208-	-209-	-210-		.071		.705	21.0	2.468
0.50				-211-	-212-	-213-	-214-	-215-	-216-		.075		.750	20.0	2.500
0.68	1.125 (28.58)	.500 (12.70)	1.325 (33.66)	-217-	-218-	-219-	-220-	-221-	-222-		.102		1.020	16.0	2.721
0.82		.500 (12.70)		-223-	-224-	-225-	-226-	-227-	-228-		.123		1.223	14.0	2.869
1.0		.562 (14.27)		-229-	-230-	-231-	-232-	-233-	-234-		.150		1.500	12.0	3.000
1.5	1.375 (34.93)	.562 (14.27)	1.575 (40.01)	-235-	-236-	-237-	-238-	-239-	-240-		.225	57.0	2.135	10.0	3.745
2.0		.670 (17.02)		-241-	-242-	-243-	-244-	-245-	-246-		.300	55.0	2.750	8.1	4.050
2.2				-247-	-248-	-249-	-250-	-251-	-252-		.330	54.0	2.970	7.5	4.121
3.0	1.875 (47.83)		2.075 (52.71)	-253-	-254-	-255-	-256-	-257-	-258-		.451	52.0	3.910	6.0	4.511
3.3				-259-	-260-	-261-	-262-	-263-	-264-		.490	50.0	4.132	5.5	4.545
4.0		.750 (19.05)		-265-	-266-	-267-	-268-	-269-	-270-		.600	48.0	4.800	4.8	4.800
4.7				-271-	-272-	-273-	-274-	-275-	-276-		.705	43.0	5.000	4.3	5.000
5.0				-277-	-278-	-279-	-280-	-281-	-282-		.750	40.0		4.0	
6.8	2.375 (60.33)	1.000 (25.40)	2.575 (65.41)	-283-	-284-	-285-	-286-	-287-	-288-		1.020	30.0		3.0	
8.0				-289-	-290-	-291-	-292-	-293-	-294-		1.200	24.0		2.4	
10.0				-295-	-296-	-297-	-298-	-299-	-300-		1.500	20.0		2.0	

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01K, CHR01L, AND CHR01M - 200 VOLTS (DC RATING)															
0.001	.562 (14.27)	.170 (4.32)	.762 (19.35)	-001-	-002-	-003-	-004-	-005-	-006-	120.0	.001	120.0	.003	80.0	.020
0.0012				-007-	-008-	-009-	-010-	-011-	-012-				.004		.024
0.0015				-013-	-014-	-015-	-016-	-017-	-018-				.005		.029
0.0018				-019-	-020-	-021-	-022-	-023-	-024-				.005		.036
0.002				-025-	-026-	-027-	-028-	-029-	-030-				.006		.040
0.0022				-031-	-032-	-033-	-034-	-035-	-036-				.007		.044
0.0027				-037-	-038-	-039-	-040-	-041-	-042-				.008		.054
0.0033				-043-	-044-	-045-	-046-	-047-	-048-				.010		.066
0.0039				-049-	-050-	-051-	-052-	-053-	-054-				.012		.070
0.0047				-055-	-056-	-057-	-058-	-059-	-060-				.014		.094
0.005				-061-	-062-	-063-	-064-	-065-	-066-		.002		.015		.100
0.0056				-067-	-068-	-069-	-070-	-071-	-072-				.017		.112
0.0068				-073-	-074-	-075-	-076-	-077-	-078-				.020		.136
0.0082				-079-	-080-	-081-	-082-	-083-	-084-				.025		.164
0.01		.193 (4.90)		-085-	-086-	-087-	-088-	-089-	-090-		.003		.030		.200
0.012		.193 (4.90)		-091-	-092-	-093-	-094-	-095-	-096-		.004		.036	78.0	.234
0.015		.235 (5.97)		-097-	-098-	-099-	-100-	-101-	-102-		.004		.045	76.0	.285
0.018	.687 (17.45)		.887 (22.53	-103-	-104-	-105-	-106-	-107-	-108-		.005		.054	74.0	.333
0.02				-109-	-110-	-111-	-112-	-113-	-114-		.006		.060	71.0	.355
0.022				-115-	-116-	-117-	-118-	-119-	-120-		.007		.066	68.0	.374
0.027		.312 (7.92)		-121-	-122-	-123-	-124-	-125-	-126-		.008		.081	65.0	.439
0.033				-127-	-128-	-129-	-130-	-131-	-132-		.010		.099	62.0	.512
0.039				-133-	-134-	-135-	-136-	-137-	-138-		.012		.118	60.0	.585
0.047				-139-	-140-	-141-	-142-	-143-	-144-		.014		.141	57.0	.670
0.050	.813 (20.65)	.312 (7.92)	1.013 (25.73)	-145-	-146-	-147-	-148-	-149-	-150-	120.0	.015	120.0	.150	56.0	.700
0.056				-151-	-152-	-153-	-154-	-155-	-156-		.017		.168	56.0	.784
0.068				-157-	-158-	-159-	-160-	-161-	-162-		.020		.204	50.0	.850
0.082				-163-	-164-	-165-	-166-	-167-	-168-		.025		.246	44.0	.902
0.10				-169-	-170-	-171-	-172-	-173-	-174-		.030		.300	42.0	1.050
0.12				-175-	-176-	-177-	-178-	-179-	-180-		.036		.360	40.0	1.201
0.15	1.063 (27.00)	.400 (10.16)	1.263 (32.08)	-181-	-182-	-183-	-184-	-185-	-186-		.045		.449	36.0	1.348

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01K, CHR01L, AND CHR01M - 200 VOLTS (DC RATING)															
0.18	1.375 (34.93)		1.575 (40.01)	-187-	-188-	-189-	-190-	-191-	-192-		.060		.600	34.0	1.700
0.22				-193-	-194-	-195-	-196-	-197-	-198-		.066		.660	32.0	1.760
0.27		.500 (12.70)		-199-	-200-	-201-	-202-	-203-	-204-		.081		.810	29.0	1.958
0.33				-205-	-206-	-207-	-208-	-209-	-210-		.099		.992	28.0	2.314
0.39				-211-	-212-	-213-	-214-	-215-	-216-		.117		1.170	27.0	2.634
0.47				-217-	-218-	-219-	-220-	-221-	-222-		.141		1.410	26.0	3.055
0.5				-223-	-224-	-225-	-226-	-227-	-228-		.150		1.500	25.0	3.125
0.56				-229-	-230-	-231-	-232-	-233-	-234-		.170	115.0	1.580	23.0	3.165
0.68		.562 (14.27)		-235-	-236-	-237-	-238-	-239-	-240-		.204	110.0	1.871	20.0	3.401
0.82	1.875 (47.83)		2.075 (52.71)	-241-	-242-	-243-	-244-	-245-	-246-		.246	100.0	2.049	19.0	3.893
1.0				-247-	-248-	-249-	-250-	-251-	-252-		.300	90.0	2.250	15.0	3.750
1.25				-253-	-254-	-255-	-256-	-257-	-258-		.370	87.0	2.701	13.0	4.050
1.5		.670 (17.02)		-259-	-260-	-261-	-262-	-263-	-264-		.449	85.0	3.184	12.0	4.494
2.0		.750 (19.05)		-265-	-266-	-267-	-268-	-269-	-270-		.600	81.0	4.050	10.0	5.000
2.5		.750 (19.05)		-271-	-272-	-273-	-274-	-275-	-276-		.750	69.0	4.320	8.2	5.125
3.0		1.000 (25.40)		-277-	-278-	-279-	-280-	-281-	-282-		.902	67.0	5.038	7.0	5.263
4.0	2.375 (60.33)	1.000 (25.40)	2.575 (65.41)	-283-	-284-	-285-	-286-	-287-	-288-		1.200	54.0	5.400	5.4	5.400
0.001	.562 (12.70)	.193 (4.90)	.762 (19.35)	-001-	-002-	-003-	-004-	-005-	-006-	240.0	.001	240.0	.006	100.0	.025
0.002	.562 (12.70)	.193 (4.90)	.762 (19.35)	-007-	-008-	-009-	-010-	-011-	-012-		.001		.012		.050
0.005	.687 (17.45)	.235 (5.97)	.887 (22.53)	-013-	-014-	-015-	-016-	-017-	-018-		.003		.030		.125
0.01	.687 (17.45)		.887 (22.53)	-019-	-020-	-021-	-022-	-023-	-024-		.006		.060		.250
0.015	.813 (20.65)		1.013(25.73)	-025-	-026-	-027-	-028-	-029-	-030-		.009		.090	94.0	.352
0.022		.312 (7.92)		-031-	-032-	-033-	-034-	-035-	-036-		.013		.132	85.0	.468
0.027				-037-	-038-	-039-	-040-	-041-	-042-		.016		.162	81.0	.547
0.033				-043-	-044-	-045-	-046-	-047-	-048-		.020		.198	78.0	.645
0.039				-049-	-050-	-051-	-052-	-053-	-054-		.023		.234	75.0	.731
0.047		.400 (10.16)		-055-	-056-	-057-	-058-	-059-	-060-		.028		.262	71.0	.834
0.05				-061-	-062-	-063-	-064-	-065-	-066-		.030		.300	70.0	.875
0.068				-067-	-068-	-069-	-070-	-071-	-072-		.041		.406	60.0	1.020
0.082	1.063(27.00)		1.253(32.08)	-073-	-074-	-075-	-076-	-077-	-078-		.049		.492	50.0	1.025
0.10	1.063(27.00)		1.253(32.08)	-079-	-080-	-081-	-082-	-083-	-084-		.060		.600	46.0	1.150

See footnotes at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses)			Dash numbers 2/						AC ratings (for sinusoidal operation from -55°C to +100°C) 3/					
	L ± .030 (0.76)	D +.020 (0.51) -.000	G max	Capacitance tolerance value (in %)						0 to .4 kHz		At 4 kHz		At 40 kHz	
				±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	Volts	Current (in A)	Volts	Current (in A)	Volts	Current (in A)
STYLES CHR01N, CHR01P, AND CHR01R - 400 VOLTS (DC RATING)															
0.15	1.375(34.93)		1.575(40.01)	-085-	-086-	-087-	-088-	-089-	-090-		.082	220.0	.824	40.0	1.498
0.22		.500 (12.70)		-091-	-092-	-093-	-094-	-095-	-096-		.132	200.0	1.100	35.0	1.925
0.33		.562 (14.27)		-097-	-098-	-099-	-100-	-101-	-102-		.198	160.0	1.322	32.0	2.645
0.47	1.875(47.83)		2.075(52.71	-103-	-104-	-105-	-106-	-107-	-108-		.282	143.0	1.680	31.0	3.643
0.50				-109-	-110-	-111-	-112-	-113-	-114-		.300	140.0	1.750	30.0	3.750
1.0		.750 (19.05)		-115-	-116-	-117-	-118-	-119-	-120-		.600	120.0	3.000	19.0	4.750
1.5	2.375(80.33)	1.000(25.40)	2.575(65.41)	-121-	-122-	-123-	-124-	-125-	-126-		.899	110.0	4.120	16.0	5.993
2.0	2.375(80.33)	1.000(15.40)	2.575(64.51)	-127-	-128-	-129-	-130-	-131-	-132-		1.200	100.0	5.000	14.0	7.000

1/ Dimensions are bare case sizes (see figure 1).
2/ The complete dash number will include the applicable letter completing the style designator (A, B, C, D, E, F, G, H, J, K, L, M, N, P, or R) as a prefix and the applicable FR level symbol as a suffix.
3/ For +125°C operation, linearly derate +100°C rating by 50 percent.

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TABLE II. Measurement values.

Measurement	CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N	CHR01B, CHR01E, CHR01H, CHR01L, and CHNR01P	CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R
Insulation resistance (terminal to terminal)			
Megohms x microfarads (min):			
At +25°C -	300,000	200,000	100,000
At +100°C -	30,000	20,000	5,000
At +125°C -	2,000	1,000	1,000
Megohms (need not exceed):			
At +25°C -	3,000,000	2,000,000	1,000,000
At +100°C -	300,000	200,000	50,000
At +125°C -	20,000	10,000	10,000
Capacitance change with temp (max)			
At -55°C -	-2.5% to 0%	-2.5% to 0%	-2.5% to 0%
At +25 (retrace) - - - - - - - - - - - - - - - -	±.25%	±.25%	±.25%
At +125°C -	-1.0% to+1.2%	-1.0% to+1.2%	±2.0%
Dissipation factor (max) - - - - - - - - - - - - -	0.15%	0.2%	0.25%

Flash point of impregnant: Not applicable.

Radiographic inspection (X-ray): Not applicable.

Thermal shock: Method 107 of MIL-STD-202, condition B.

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIA; no evidence of leakage with the exception of the following:

Styles	lbf/in^2	Exposure time hours	Maximum dwell hours	R1 Reject limit ($atm\ cm^3/s$)
CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, and CHR01P	45 ± 5	1 +1-0	1	1×10^{-6}
CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R	45 ± 5	1 +1-0	1	1×10^{-5}

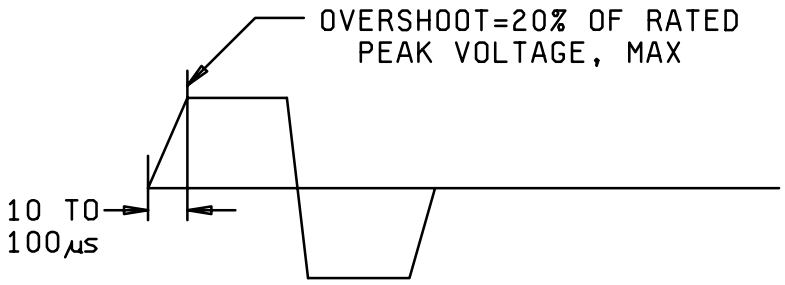
Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202; no momentary breakdown:

Terminal to terminal:

AC: 100 ± 10 Hz square wave, peak-to-peak voltage, three times dc rated voltage for 60 seconds, minimum. Need not exceed 800 peak-to-peak.

DC: 200 percent of dc rated voltage for 60 seconds, min.

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Insulation resistance (IR): Method 302 of MIL-STD-202. Charge time 5 minutes, maximum; however, for capacitance values greater than 1.0 µF, an additional 1 minute per µF is permitted.

Terminal to terminal (dc rated voltage): See table II.

Terminals to case: 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Test frequency: 1 kHz ±100 Hz.

Limit of accuracy: Within ± 0.05 percent.

Dissipation factor (DF): See table II.

Test frequency: 1 kHz ± 100 Hz (capacitance values ≤ 1.0 µF) or 100 ± 10 Hz (capacitance values > 1.0 µF).

Limit of accuracy: Within ± 0.05 percent or within 10 percent of table II value.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).

125 percent of dc rated voltage applied. See MIL-PRF-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G).

50 percent of dc rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: 5 percent.

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Insulating sleeves: Greater than 4,000 V dc.

Terminal to terminal: 200 percent of dc rated voltage.

Terminals to case: 200 percent of dc rated voltage.

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IR:

Insulating sleeves: 1,000 megohms, minimum.

Terminal to terminal: Not less than 50 percent of value specified in table II.

Terminals to case: 5,000 megohms, minimum.

Capacitance:

CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N: ±0.25 percent, maximum.

CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P: ±0.5 percent, maximum.

CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: ±1.0 percent, maximum.

DF: See table II.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I.

50 percent of dc rated voltage applied. No arcing or other evidence of electrical breakdown.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, ΔCapacitance, and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.

Condition A: Applied force 5 pounds.

Condition D: 3 rotations of 360 degrees.

Low temperature and capacitance change with temperature:

Low temperature: In accordance with MIL-PRF-39022.

Capacitance change with temperature: In accordance with MIL-PRF-39022, except measurements may be made at -50°C +0°C, -3°C; -25°C ±3°C; 0 ±3°C; +25°C ±3°C; +50°C ±3°C; +75°C ± 3°C; +100 ±3°C; and +125°C ±3°C.

ΔCapacitance (at -55°C, +25°C (retrace), and +125°C): See table II.

Fault count: Not applicable.

Fungus: Method 508 of MIL-STD-810.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition C, except the time shall be 10 seconds ± 1 second.

IR: See table II.

ΔCapacitance: Within ± 0.25 percent of initial value.

DF: See table II.

Resistance to solvents: Method 215 of MIL-STD-202.

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Life (at +100°C): Method 108 of MIL-STD-202.

Accelerated conditions: 140 percent of dc rated voltage for 2,000 +72, -0 hours.

Rated conditions: 100 percent of dc rated voltage for 10,000 +96, -0 hours.

DF (at +100°C) between 24 and 48 hours of test: See table II.

IR:

Insulating sleeves: 1,000 megohms, minimum.

Terminal to terminal: Not less than 33.3 percent of initial requirement.

Terminals to case: 5,000 megohms, minimum.

ΔCapacitance:

CHR01A, CHR01D, CHR01G, CHR01K, and CHR01N: ± 2.0 percent of initial value.

CHR01B, CHR01E, CHR01H, CHR01L, and CHR01P: ± 4.0 percent of initial value.

CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: ± 5.0 percent of initial value.

DF (at +25°C)after life:

CHR01A, CHR01B, CHR01D, CHR01E, CHR01G, CHR01H, CHR01K, CHR01L, CHR01N, and CHR01P: 0.25 percent, maximum.

CHR01C, CHR01F, CHR01J, CHR01M, and CHR01R: 0.35 percent, maximum.

AC life: 140 percent of ac rated voltage for 240 +48, -0 hours. Test samples shall be in accordance with table III.

TABLE III. AC life test samples.

DC rated voltage (volts)	AC rating at 400 Hz (volts)	Life test voltage RMS	Life test voltage frequency	Total capacitance value of sample (µF)	Capacitance range for selection of sample
30	22	30	4 kHz	40	.001 to 4 µF
50	36	50	4 kHz	40	.001 to 4 µF
100	60	84	4 kHz	20	.001 to 2.2 µF
200	120	168	400 Hz	40	.001 to 10 µF
400	240	336	400 Hz	20	.001 to 2 µF

NOTE: Cross-reference of substitute items: In accordance with table IV.

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TABLE IV. Cross reference of substitute items.

Superseded PIN	Replacement PIN	Superseded PIN	Replacement PIN	Superseded PIN	Replacement PIN
STYLES CHR01A, CHR01B, AND CHR01C - 30 VOLTS					
-265-	-319-	-277-	-331-	-289-	-343-
-266-	-320-	-278-	-332-	-290-	-344-
-267-	-321-	-279-	-333-	-291-	-345-
-268-	-322-	-280-	-334-	-292-	-346-
-269-	-323-	-281-	-335-	-293-	-347-
-270-	-324-	-282-	-336-	-294-	-348-
-271-	-325-	-283-	-337-	295-	-349-
-272-	-326-	-284-	-338-	-296-	-350-
-273-	-327-	-285-	-339-	-297-	-351-
-274-	-328-	-286-	-340-	-298-	-352-
-275-	-329-	-287-	-341-	-299-	-353-
-276-	-330-	-288-	-342-	-300-	-354-
STYLES CHR01D, CHR01E, AND CHR01F - 50 VOLTS					
-247-	-301-	-259-	-313-	-271-	-325-
-248-	-302-	-260-	-314-	-272-	-326-
-249-	-303-	-261-	-315-	-273-	-327-
-250-	-304-	-262-	-316-	-274-	-328-
-251-	-305-	-263-	-317-	-275-	-329-
-252-	-306-	-264-	-318-	-276-	-330-
-253-	-307-	-265-	-319-	277-	-331-
-254-	-308-	-266-	-320-	-278-	-332-
-255-	-309-	-267-	-321-	-279-	-333-
-256-	-310-	-268-	-322-	-280-	-334-
-257-	-311-	-269-	-323-	-281-	-335-
-258-	-312-	-270-	-324-	-282-	-336-

Marking: In accordance with MIL-PRF-39022 and the following:

Part or Identifying Number (PIN): Consists of the basic number of this specification sheet with a dash number coded as shown in table I.

EXAMPLE OF MARKING:

M39022/9
A001LJAN: PIN and "JAN" Marking.
7304AB: Date code and lot symbol.
TMXXXX: Trademark and source code.

Packaging: Capacitors shall be furnished in tape and reel packaging when so specified in the ordering data.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

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